

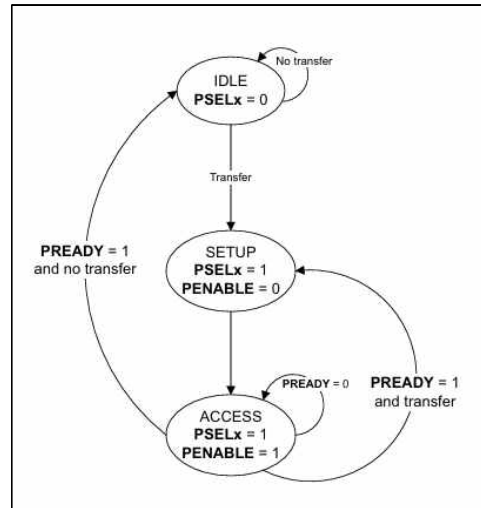
APB (Advanced Peripheral Bus)

non-pipelined, synchronous. low-cost interface, minimal power consumption

APB bridge = Requester

APB peripherals = Completer

• State Diagram

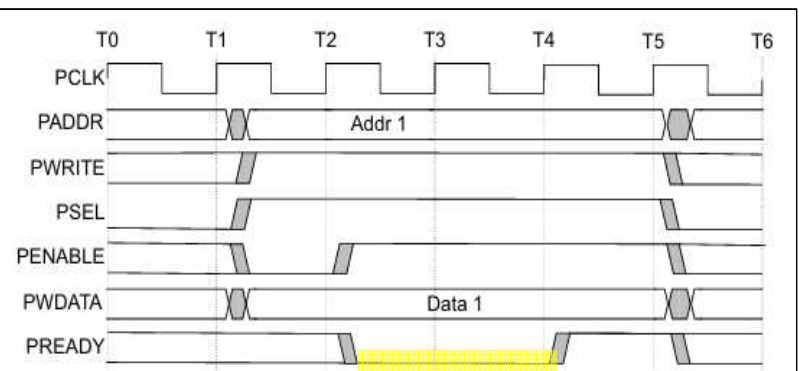
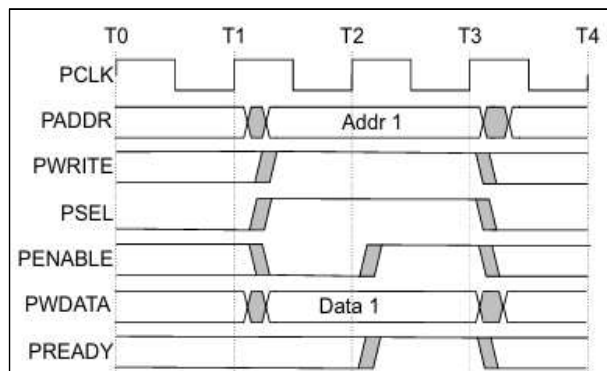


2-cycle

1. IDLE : default
2. SETUP : PSEL = 1 , 다음 clock의 rising edge에서 항상 ACCESS상태로 감.
3. ACCESS (ENABLE) : PENABLE = 1 , transition동안 address, write, select signal이 stable해야함.
다음 transfer의 존재 여부에 따라 SETUP 혹은 IDLE로 이동

• Transfers

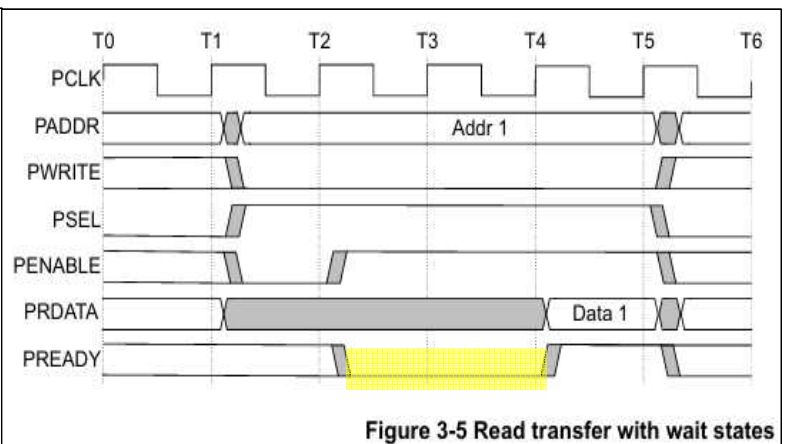
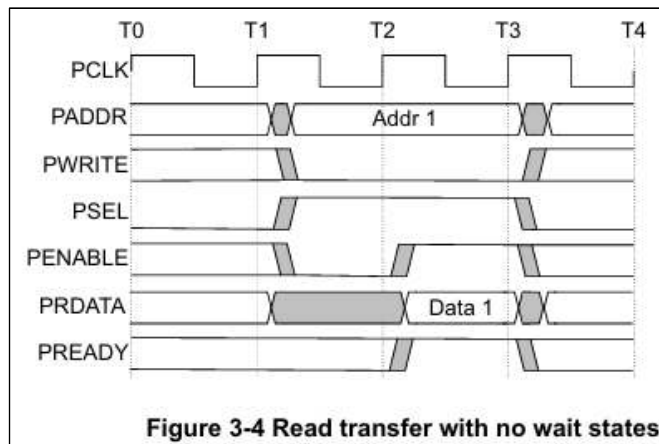
1. Write transfer



*PADDR, PWRITE, PWDATA, PSEL은 PREADY가 1일 때까지 변경 불가

- PADDR, PWRITE, PWDATA, PSEL은 setup에서 바뀜. (ACCESS에서 유지)
- 같은 slave → PSEL유지, 다른 slave → PSEL=0

2. Read transfer

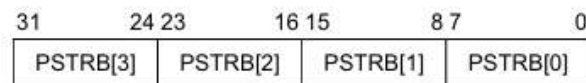


*write와 data방향만 다르고 타이밍 동일함.

- PRDATA는 ENABLE cycle 동안 valid.
- data는 마지막 rising edge에서 read.

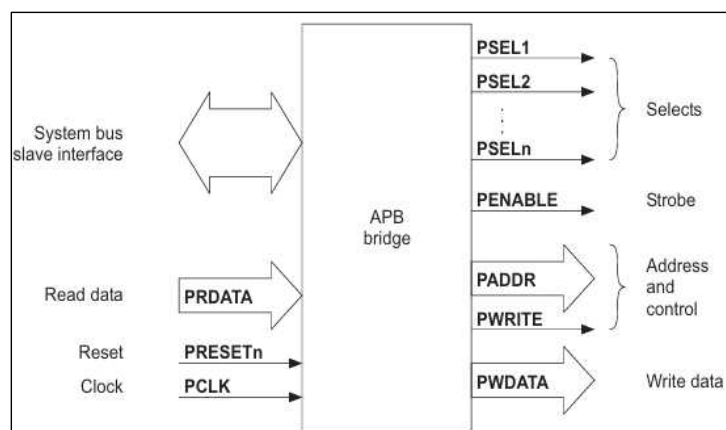
• Optional signal

1. PSTRB : 부분쓰기 기능. 8bit마다 하나씩 있음. ($PWDATA[(8n + 7):(8n)]$)



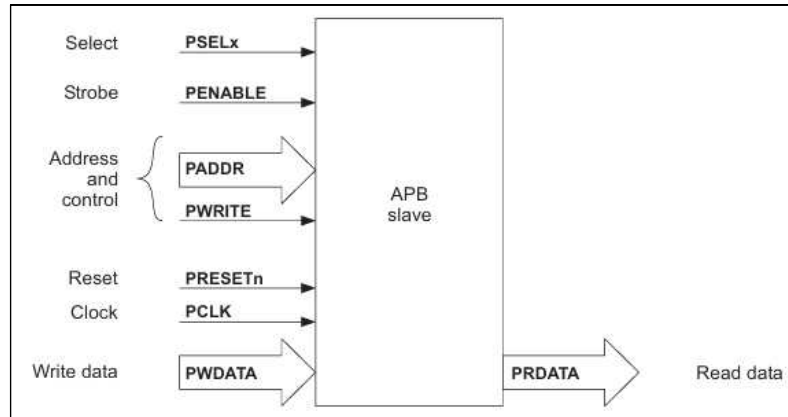
2. PSLVERR : slave 동작 error검출 신호. last cycle of APB transfer (PSEL, PENABLE, PREADY = 1)일 때 동작.
3. PPROT [2:0] : protection unit
([0]: normal or privileged / [1]: Secure or Non-secure/ [2]: Data or Instruction)
4. RME : 보안 영역 확장. PPROT[1]과 조합 (Secure, Non-secure, **Root**, **Realm**)
5. PWAKEUP : APB interface를 사용하기위해 쓰는 signal
6. User signaling : PAUSER, PWUSER, PRUSER, PBUSER

• APB bridge



- 2cycle동안 주소 유지
- transfer동안 하나의 select signal만 가능함.
- write transfer : data onto APB
- read transfer : data onto the system bus

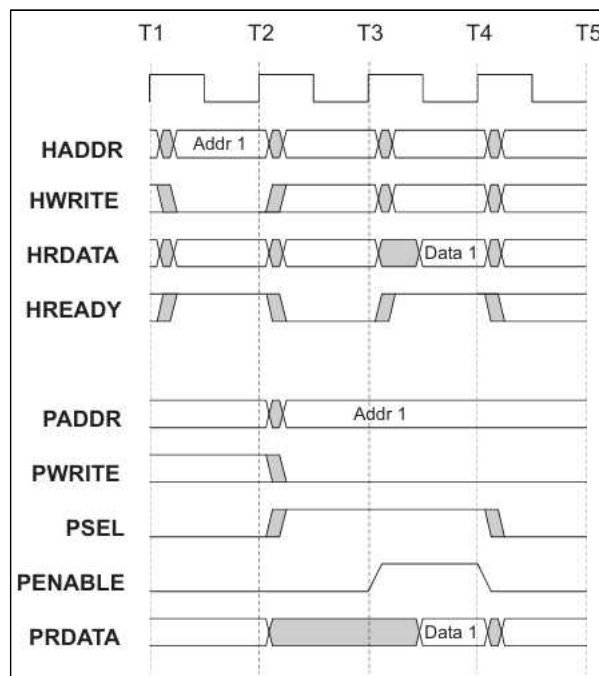
- APB slave



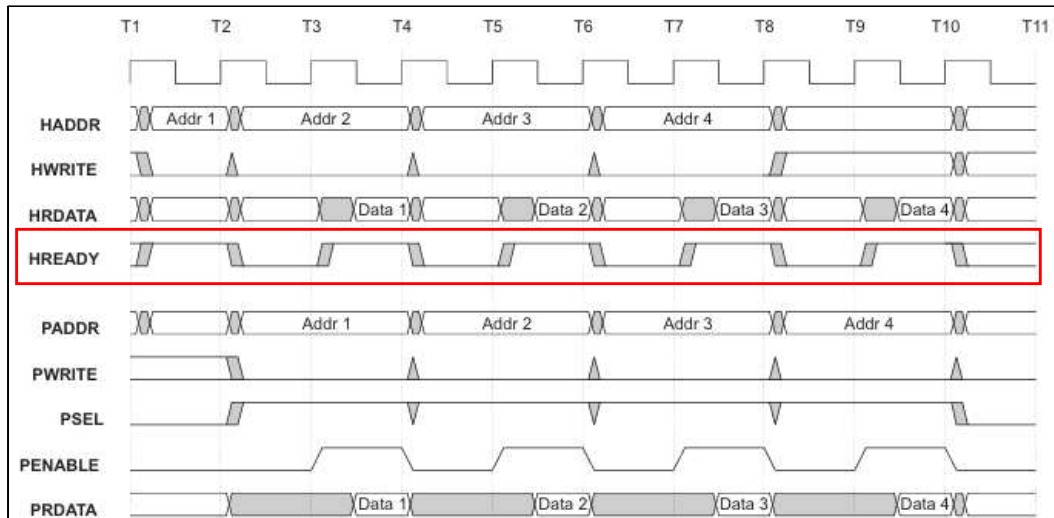
- write transfer : PSEL = 1일 때 1. rising edge of PCLK 2. rising edge of PENABLE에서 data latch
- read transfer : PSEL = 1 & PENABLE = 1 & PWRITE = 0

- Interfacing APB to AHB

1. read transfer



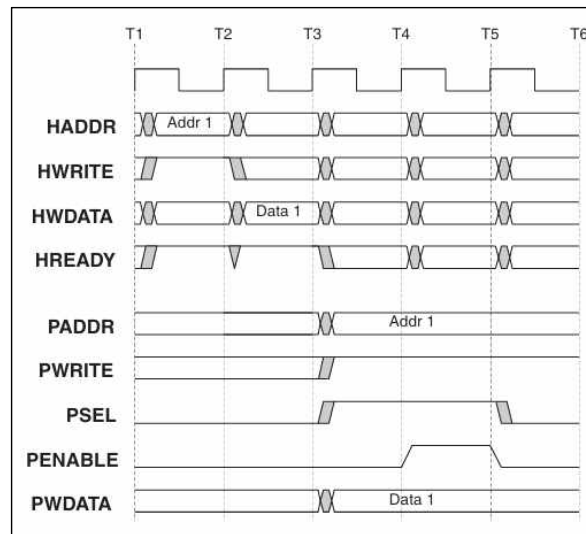
- T2 : Bridge에서 addr sample (select signal형성)
- PRDATA는 AHB로 data작업 전달. T4의 rising edge에서 data sampling
- if) clk frequency 빠르면 ENABLE 사이클 끝에서 rdata를 레지스터에 저장하고 다음 사이클에서 이 data를 AHB 버스 마스터로 전달 → 더높은 clock frequency



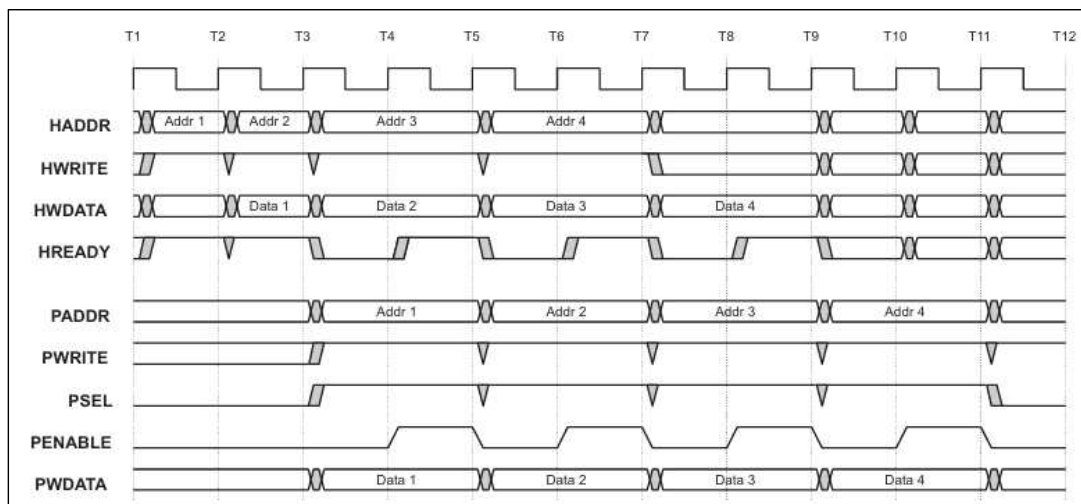
[burst read]

모든 read 전송은 하나의 wait state가 필수 → HREADY가 한 번 내려감

2. Write transfer



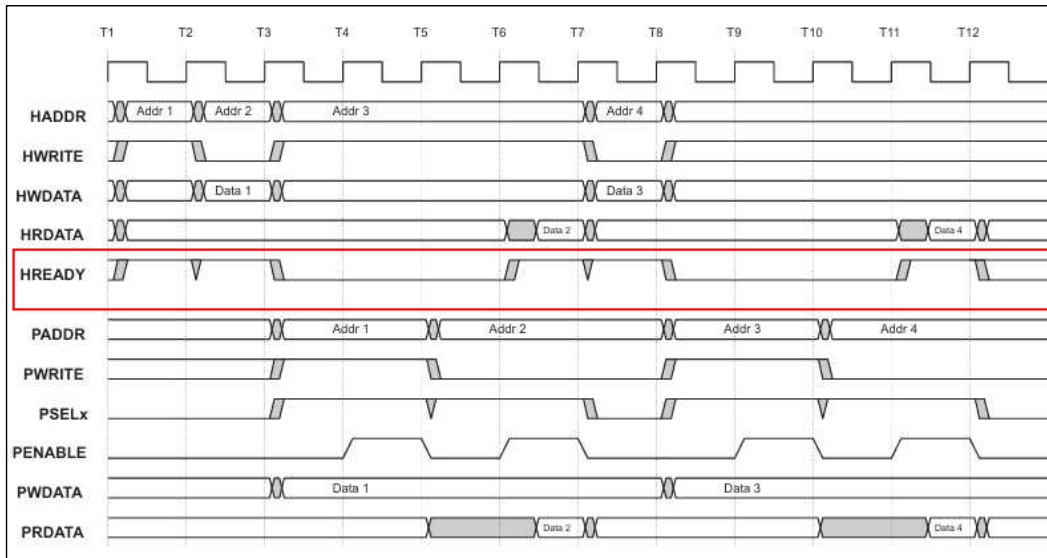
- single write일 땐 wait state 불필요
- Address와 data는 bridge 전송 중 유지해야함.



[write burst]

- wait state 1-cycle 필요.
- bridge는 address register 2개 필요. (전송이 진행되는 동안 다음 주소 sampling)

3. Back to back Transfers



[read transfer가 write transfer 바로 뒤에 오는 경우]

- 3 wait state 필요

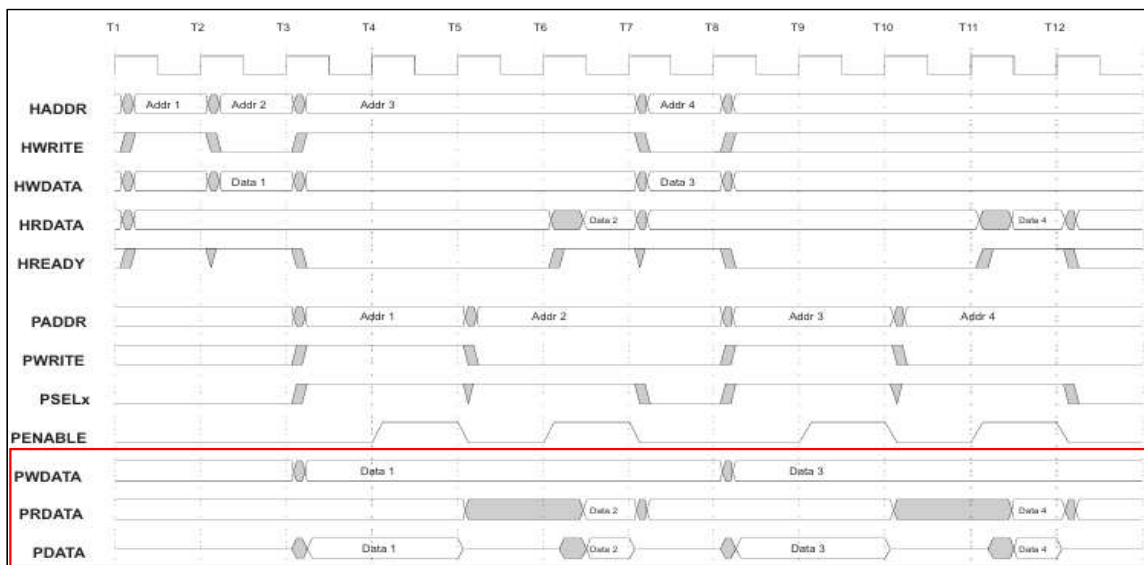
write : master → slave

read : slave → master

write setup → write enable → read setup → read enable → AHB

• Tristate data bus implementations

: read와 write를 하나의 버스로 합치는 것 (read와 write는 동시에 발생하지 않기 때문)



- PWDATA : Master → Slave (Write)

- PRDATA : Slave → Master (Read)

SETUP cycle 동안과 Idle 상태에서는 데이터 버스가 tristate 상태가 되므로, 서로 다른 드라이버가 버스를 구동하기 전에 항상 1클럭의 turnaround time이 생김.